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University ST4005CEM Database System
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1. Write a CREATE TABLE statement for the CUSTOMER table with customer_id primary key, name, address, phone, and email columns

```
1  
2  
3  
4  
5 • create table customer (  
6     customer_id int primary key,  
7     name varchar(20),  
8     address varchar(20),  
9     phone char(10),  
10    email varchar(30)  
11  
12    );
```

2. Write a CREATE TABLE statement for the ORDER table that includes a foreign key referencing CUSTOMER customer_id.

```
13  
14 • create table orders (  
15     order_id int primary key,  
16     customer_id int,  
17     status enum("In Transit","Delivered","Cancelled"),  
18     order_date date default(curdate()),  
19     foreign key (customer_id) references customer(customer_id)  
20 );
```

3. Create the SHIPMENT table with shipment_id as primary key and route_id as a foreign key referencing ROUTE route_id.

```
29 • create table shipment(  
30     shipment_id int primary key,  
31     order_id int,  
32     route_id int,  
33     status enum("In Transit","Delivered","Cancelled"),  
34     shipment_date date default (curdate()),  
35     foreign key (order_id) references orders(order_id),  
36     foreign key (route_id) references route(route_id)  
37 );
```

4. Create the TRACKING table with tracking_id PK, shipment_id FK, location string, and timestamp date.

```
39 • create table tracking(  
40     tracking_id int primary key,  
41     shipment_id int,  
42     location varchar(20),  
43     timestamp date ,  
44     foreign key (shipment_id) references shipment(shipment_id)  
45 );
```

5. Insert a new customer with name 'Asha Thapa', address 'Kathmandu', phone '9800000001', and email 'asha@mail.com' .

```
14 • insert into customer values  
15     (1,"Asha Thapa","Kathmandu","9800000001", "asha@mail.com");  
16  
17 • select * from customer;  
18 • create table orders (
```

Result Grid

	customer_id	name	address	phone	email
▶	1	Asha Thapa	Kathmandu	9800000001	asha@mail.com
*	NULL	NULL	NULL	NULL	NULL

6. Insert an order for customer_id=1, status='Pending', and today's date as order_date.

```
26 • alter table orders modify status enum("In Transit","Delivered","Cancelled","Pending");  
27 • insert into orders (order_id,customer_id) values(1,1,"Pending");  
28  
29 • select * from orders;
```

Result Grid

	order_id	customer_id	status	order_date
▶	1	1	Pending	2026-05-06
*	NULL	NULL	NULL	NULL

7. Insert a ROUTE record with origin='Kathmandu', destination='Pokhara', distance=200.

```
36 • insert into route values(1,"Kathmandu","Pokhara",200);
37 • select * from route;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

	route_id	origin	destination	distance
▶	1	Kathmandu	Pokhara	200
*	NULL	NULL	NULL	NULL

8. Insert a SHIPMENT record that links order_id=3 with route_id=1, status='In Transit', and today's date.

```
51 • insert into shipment (shipment_id,order_id,route_id,status) values(1,3,1,"In Transit");
52 • select * from shipment;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	shipment_id	order_id	route_id	status	shipment_date
▶	1	3	1	In Transit	2026-05-06
*	NULL	NULL	NULL	NULL	NULL

9. Retrieve all columns for every customer in the CUSTOMER table.

```
18 • select * from customer;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

	customer_id	name	address	phone	email
▶	1	Asha Thapa	Kathmandu	9800000001	asha@mail.com
	2	Nischal Shrestha	Bratchowk	9800000002	nischal@gmail.com
	3	Samir DC	Chovar	9800000003	samir@gmail.com
*	NULL	NULL	NULL	NULL	NULL

10. Display the order_id, status, and order_date for all orders that have status 'Delivered'.

```
63 • select * from orders where status="Delivered"
```

Result Grid | Filter Rows: | Edit: | Export/Import:

	order_id	customer_id	status	order_date
▶	3	1	Delivered	2026-05-06
*	NULL	NULL	NULL	NULL

11. List all shipments sorted by shipment_date in descending order.

```
64 • select * from shipment order by shipment_date desc;
```

	shipment_id	order_id	route_id	status	shipment_date
▶	1	3	1	In Transit	2026-05-06
	2	2	1	Delivered	2026-05-06
	4	2	1	In Transit	2026-05-04
	3	1	1	In Transit	2026-05-03
*	NULL	NULL	NULL	NULL	NULL

shipment 21 x

12. Retrieve the top 5 heaviest parcels (by weight) from the PARCEL table.

```
81 • select * from parcel order by weight desc limit 5;
```

	parcel_id	order_id	parcel_type	weight	dimensions
▶	5	2	Oversized	12.4	60x50x40
	10	1	Oversized	9.7	55x45x35
	3	3	Fragile	5	40x30x25
	7	1	Fragile	4.2	45x35x30
	4	1	Standard	3.8	35x25x20
*	NULL	NULL	NULL	NULL	NULL

13. Find all customers whose name starts with the letter 'A'.

```
86 • select * from customer where name like "a%";
```

	customer_id	name	address	phone	email
▶	1	Asha Thapa	Kathmandu	9800000001	asha@mail.com
*	NULL	NULL	NULL	NULL	NULL

14. Retrieve all customers whose email ends with '@gmail.com'.

```
87 • select * from customer where email like "%@gmail.com";
```

Result Grid | Filter Rows: | Edit: | Export/Import: |

	customer_id	name	address	phone	email
▶	2	Nischal Shrestha	Bratchowk	9800000002	nischal@gmail.com
	3	Samir DC	Chovar	9800000003	samir@gmail.com
*	NULL	NULL	NULL	NULL	NULL

15. Add a new column called 'loyalty_points' of type INT to the CUSTOMER table

```
88 • alter table customer add column loyalty_points int;  
89 • select * from customer;  
90
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wra

	customer_id	name	address	phone	email	loyalty_points
▶	1	Asha Thapa	Kathmandu	9800000001	asha@mail.com	NULL
	2	Nischal Shrestha	Bratchowk	9800000002	nischal@gmail.com	NULL
	3	Samir DC	Chovar	9800000003	samir@gmail.com	NULL
*	NULL	NULL	NULL	NULL	NULL	NULL

16. Rename the column 'status' in the ORDER table to 'order_status'

```
91 • alter table `orders` rename column status to order_status;  
92 • select * from orders;  
93
```

Result Grid | Filter Rows: | Edit: | Export/Import: |

	order_id	customer_id	order_status	order_date
▶	1	1	Pending	2026-05-06
	2	1	In Transit	2026-05-06
	3	1	Delivered	2026-05-06
*	NULL	NULL	NULL	NULL

17. Change the data type of the 'distance' column in ROUTE from INT to FLOAT

```
94 • alter table route modify distance float;
95 • desc route;
```

Result Grid | Filter Rows: | Export: | Wrap C

	Field	Type	Null	Key	Default	Extra
▶	route_id	int	NO	PRI	NULL	
	origin	varchar(20)	YES		NULL	
	destination	varchar(20)	YES		NULL	
	distance	float	YES		NULL	

18. Delete all orders that have a status of 'Cancelled'

```
96
97 • delete from orders where order_status="cancelled";
98
99 • set sql_safe_updates=0;
```

Output :



Action Output

	#	Time	Action
✖	17	17:25:59	set safe_updates=0
✔	18	17:26:05	set sql_safe_updates=0
✔	19	17:26:07	delete from orders where order_status="cancelled"

19. Update the status of order_id=3 to 'Delivered'.

```

99 • update orders set order_status="Delivered" where order_id=3;
100 • set sql_safe_updates=0;
101

```

Result Grid				
Filter Rows:				
	order_id	customer_id	order_status	order_date
▶	1	1	Pending	2026-05-06
	2	1	In Transit	2026-05-06
	3	1	Delivered	2026-05-06
•	NULL	NULL	NULL	NULL

20. Increase the weight of all parcels of type 'box' by 0.5 k

```

104
105 • update parcel set weight=0.5 where parcel_type="box";

```

Output			
Action Output			
#	Time	Action	
✓ 28	17:31:20	update parcel set weight=0.5 where parcel_type="express"	
✓ 29	17:31:22	select * from parcel LIMIT 0, 1000	
✓ 30	17:31:30	update parcel set weight=0.5 where parcel_type="box"	

21. Update the phone number for customer_id=3 to '9811223344'

```

105 • select * from customer;
106 • update customer set phone="9811223344" where customer_id=3;

```

Result Grid						
Filter Rows:						
	customer_id	name	address	phone	email	loyalty_points
▶	1	Asha Thapa	Kathmandu	9800000001	asha@mail.com	NULL
	2	Nischal Shrestha	Bratchowk	9800000002	nischal@gmail.com	NULL
	3	Samir DC	Chovar	9811223344	samir@gmail.com	NULL
•	NULL	NULL	NULL	NULL	NULL	NULL

22. Show all shipments along with the route origin and destination for each

```
108 • select *,origin from shipment join route on shipment.route_id=route.route_id;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	shipment_id	order_id	route_id	status	shipment_date	route_id	origin	destination	distance	origin
	1	3	1	In Transit	2026-05-06	1	Kathmandu	Pokhara	200	Kathmandu
	2	2	1	Delivered	2026-05-06	1	Kathmandu	Pokhara	200	Kathmandu
	3	1	1	In Transit	2026-05-03	1	Kathmandu	Pokhara	200	Kathmandu
	4	2	1	In Transit	2026-05-04	1	Kathmandu	Pokhara	200	Kathmandu

23. List every order along with the customer's name and email who placed it

```
110 • select *,name,email from orders join customer on orders.customer_id=customer.customer_id;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	order_id	customer_id	order_status	order_date	customer_id	name	address	phone	email	loyalty_points	name	email
▶	1	1	Pending	2026-05-06	1	Asha Thapa	Kathmandu	9800000001	asha@mail.com	NULL	Asha Thapa	asha@mail.com
	2	1	In Transit	2026-05-06	1	Asha Thapa	Kathmandu	9800000001	asha@mail.com	NULL	Asha Thapa	asha@mail.com
	3	1	Delivered	2026-05-06	1	Asha Thapa	Kathmandu	9800000001	asha@mail.com	NULL	Asha Thapa	asha@mail.com

24. List customers who have never placed an order

```
114 • SELECT *
115 FROM customer
116 LEFT JOIN orders ON customer.customer_id = orders.customer_id
117 WHERE orders.customer_id IS NULL;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	customer_id	name	address	phone	email	loyalty_points	order_id	customer_id	order_status	order_date
▶	2	Nischal Shrestha	Bratchowk	9800000002	nischal@gmail.com	NULL	NULL	NULL	NULL	NULL
	3	Samir DC	Chovar	9811223344	samir@gmail.com	NULL	NULL	NULL	NULL	NULL